Subject Code: BP304T

PAPER ID-311528

Roll No: | | | | | | | | | | | | | |

# BPHARM
## (SEM III) THEORY EXAMINATION 2024-25
## PHARMACEUTICAL ENGINEERING

**TIME: 3 HRS** **M.MARKS: 75**

**Note:** **1.** Attempt all Sections. If require any missing data; then choose suitably.
**2.** Any special paper specific instruction.

### SECTION A

**1. Attempt *all* questions in brief.** **10 x 2 = 20**

| | |
|---|---|
| a. | What is critical speed of a ball mill. |
| b. | Define Reynold number. |
| c. | What is convection? |
| d. | State Bernoulli's theorem |
| e. | How does tray dryer works in a pharma industry? |
| f. | Justify with examples the differences between solid and liquid mixing . |
| g. | What is filter aid and filter media? |
| h. | What is grey body and black body? |
| i. | What is the difference between Sedimentation & Elutriation? |
| j. | What is galvanic corrosion? |

### SECTION B

**2. Attempt any *two* parts of the following:** **2 x 10 = 20**

| | |
|---|---|
| a. | Suggest the factors affecting size reduction. Explain with a diagram the working principles, applications and process of size reduction of ball mill. |
| b. | Narrate the various types of corrosion. Suggest some measures to control such corrosion |
| c. | Draw and explain drying rate curve. Write a short note on fluidized bed dryer |

### SECTION C

**3. Attempt any *five* parts of the following:** **7 x 5 = 35**

| | |
|---|---|
| a. | What is filter aid? Explain the various factors influencing filtration. |
| b. | Discuss the working uses, merits and demerits of Perforated basket centrifuge, Non-perforated basket centrifuge |
| c. | What are the various processes of heat transfer ? Explain the various types of shell and tube heat exchangers with the help of diagrams. |
| d. | Explain the principles, construction and uses of twin shell blender. |
| e. | What is filter aid? Explain the various factors influencing filtration. |
| f. | Explain steam and molecular distillation alongwith its pharmaceutical applications |
| g. | Write short notes on:<br>i. Sieve shaker ii. Pitot tube |

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